

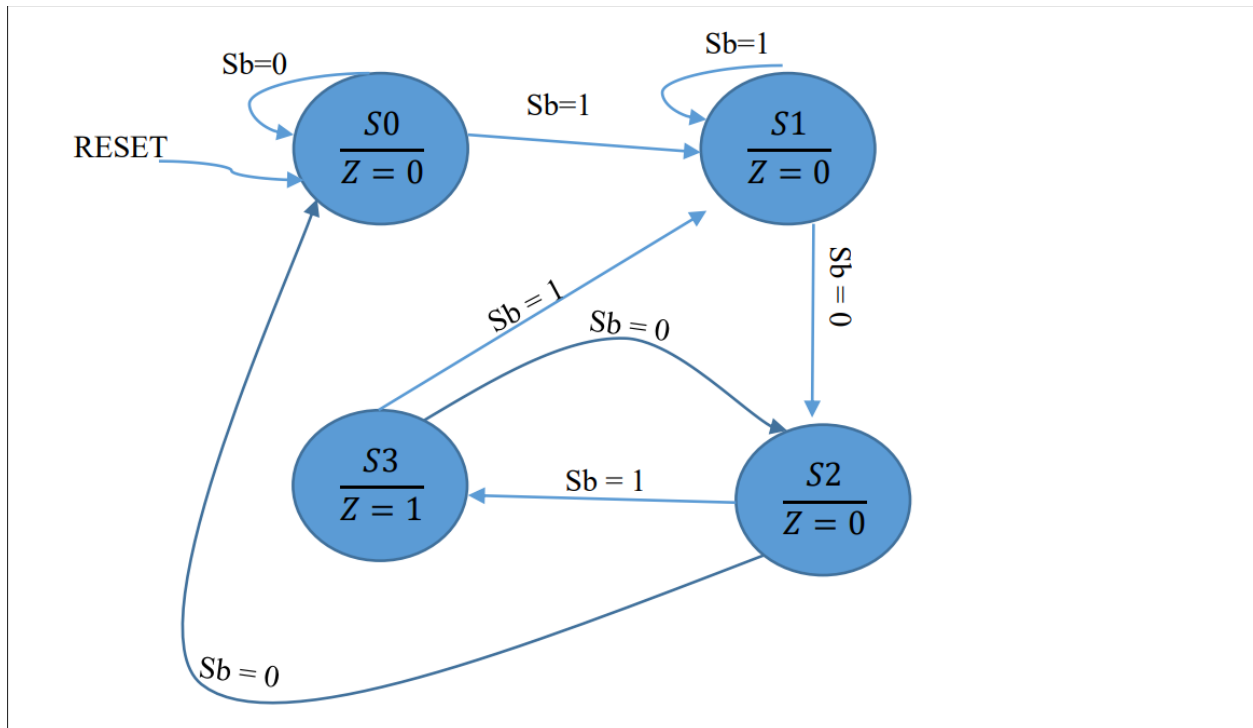
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Ex01:

State diagram:



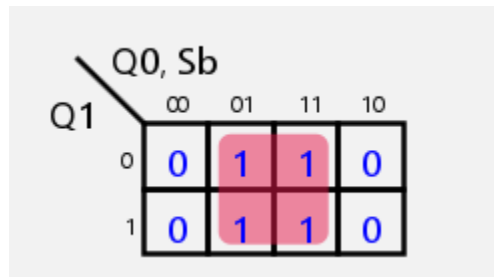
State table:

Current State	Next State		Output Z
	Sb = 0	Sb = 1	
S0	S0	S1	0
S1	S2	S1	0
S2	S0	S3	0
S3	S2	S1	1

State	Q1Q0
S0	00
S1	01
S2	10
S3	11

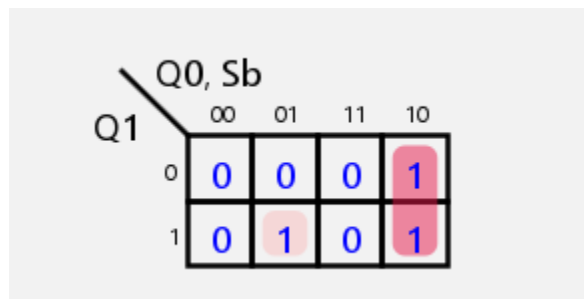
Current State Q1Q0	Next State Q1Q0		D1D0		Z
	Sb = 0	Sb = 1	Sb = 0	Sb = 1	
00	00	01	00	01	0
01	10	01	10	01	0
10	00	11	00	11	0
11	10	01	10	01	1

K-map (D0):



$$D0 = Sb$$

K-map (D1):

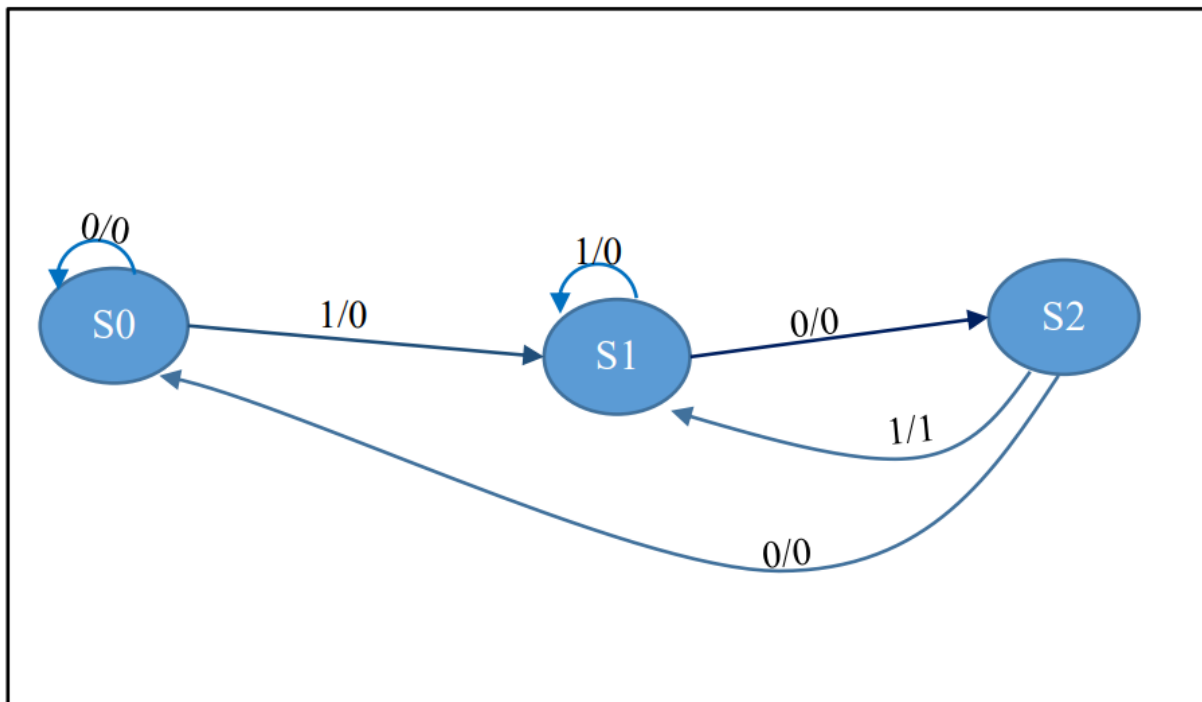


$$D1 = Q0 \cdot \sim Sb + Q1 \cdot \sim Q0 \cdot Sb$$

$$Z = Q1 \cdot Q0$$

Ex02:

State diagram:



State table:

State	Q1Q0
S0	00
S1	01
S2	10

Sb	Current State	Next State	Z
0	S0	S0	0
1	S0	S1	0
0	S1	S2	0
1	S1	S1	0
0	S2	S0	1
1	S2	S1	0

Sb	Current State Q1Q0	Next State Q1Q0	D1D0	Z
0	00	00	00	0
1	00	01	01	0
0	01	10	10	0
1	01	01	01	0
0	10	00	00	0
1	10	01	01	1

K-map (D0):

Q0, Sb		00	01	11	10
Q1	0	0	1	1	0
	1	0	1	-	-

$$D0 = Sb$$

K-map (D1):

Q0, Sb		00	01	11	10
Q1	0	0	0	0	1
	1	0	0	-	-

$$D1 = Q0 \cdot \sim Sb$$

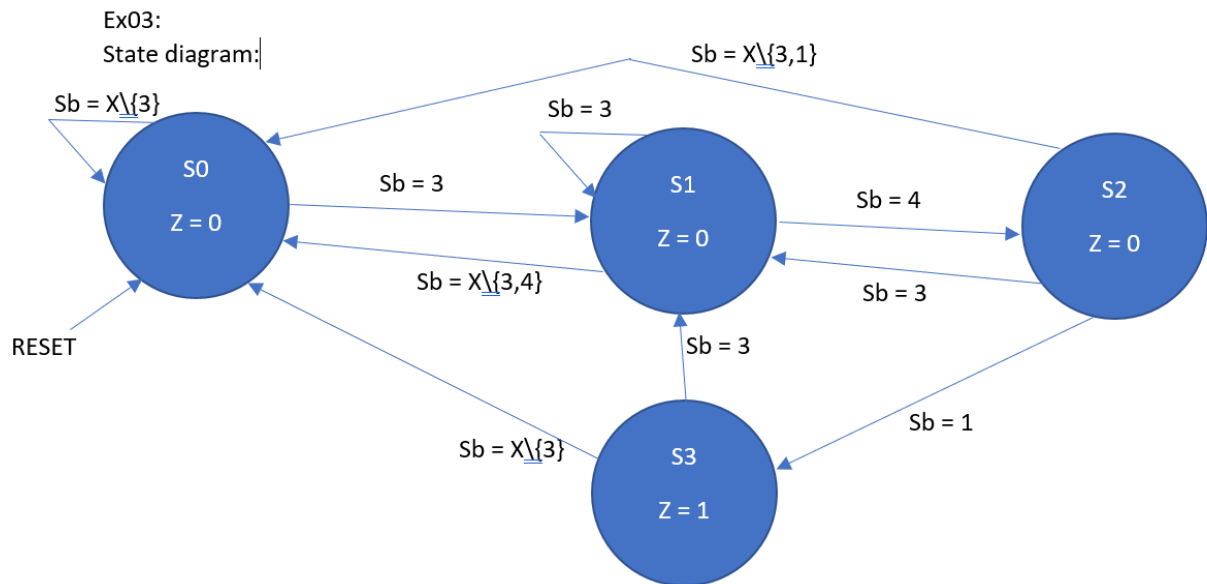
K-map (Z):

Q0, Sb		00	01	11	10
Q1	0	0	0	0	0
	1	0	1	-	-

$$Z = Q1 \cdot Sb$$

Ex03:

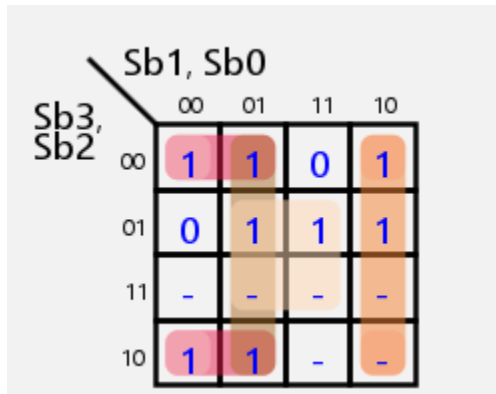
State diagram:



Encoder table:

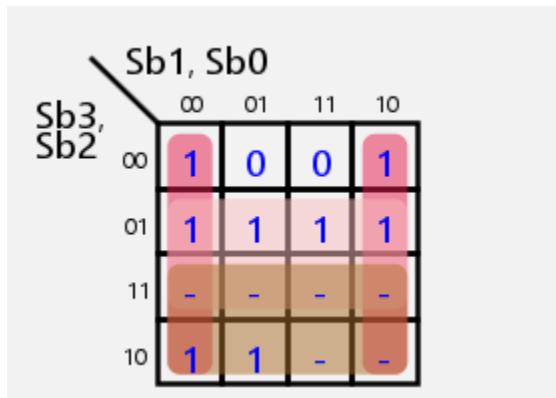
Sb (Binary)	Sb (Binary) Sb3Sb2Sb1Sb0	I1I0
0	0000	11
1	0001	10
2	0010	11
3	0011	00
4	0100	01
5	0101	11
6	0110	11
7	0111	11
8	1000	11
9	1001	11

K-map (I1):



$$I1 = \sim Sb2 \cdot \sim Sb1 + \sim Sb2 \cdot \sim Sb0 + \sim Sb3 \cdot \sim Sb1 \cdot Sb0 + Sb1 \cdot \sim Sb0 + \sim Sb3 \cdot Sb2 \cdot Sb0 + Sb3 \cdot \sim Sb2 + Sb2 \cdot Sb1$$

K-map(I0):



$$I0 = \sim Sb0 + Sb2 + Sb3$$

I1I0	Current State Q1Q0	Next State Q1Q0	Z
00	00	01	0
01	00	00	0
10	00	00	0
11	00	00	0
00	01	01	0
01	01	10	0
10	01	00	0
11	01	00	0
00	10	01	0
01	10	00	0
10	10	11	0
11	10	00	0
00	11	01	1
01	11	00	1
10	11	00	1
11	11	00	1

K-map (D1):

Q1, Q0					
		00	01	11	10
I1, I0	00	0	0	0	0
	01	0	1	0	0
	11	0	0	0	0
	10	0	0	0	1

$$D1 = \sim I1 \cdot I0 \cdot \sim Q1 \cdot Q0 + I1 \cdot \sim I0 \cdot Q1 \cdot \sim Q0$$

K-map (D0):

Q1, Q0					
		00	01	11	10
I1, I0	00	1	1	1	1
	01	0	0	0	0
	11	0	0	0	0
	10	0	0	0	1

$$D0 = \sim I1 \cdot \sim I0 + \sim I0 \cdot Q1 \cdot \sim Q0$$

K-map (Z):

Q1, Q0					
		00	01	11	10
I1, I0	00	0	0	1	0
	01	0	0	1	0
	11	0	0	1	0
	10	0	0	1	0

$$Z = Q_1 \cdot Q_0$$